



6.3.2 巴特沃思低通逼近

令 $H_{a_N}(s)$ 代表归一化系统的系统函数， $H_a(s)$ 代表截止频率为 Ω_c' 的低通系统的传递函数，那么归一化系统函数中的变量 s 用 s/Ω_c' 代替后，就得到所需滤波器的系统函数 $H_a(s)$ ，即：

$$s \rightarrow \frac{s}{\Omega_c'} \quad (5-18)$$

$$H_a(s) = H_{a_N} \left[\frac{s}{\Omega_c'} \right] \quad (5-19)$$



表 5-1 巴特沃思归一化低通滤波器分母多项式

$s^N + a_{N-1}s^{N-1} + \dots + a_2s^2 + a_1s + 1$ 的系数 ($a_1 = a_N = 1$)

N	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9
1	1								
2	1.4142436								
3	2.0000000	2.0000000							
4	2.6131259	3.4142136	2.6131259						
5	3.2360680	5.2360680	5.2360680	3.2360680					
6	3.8637033	7.4641016	9.1416202	7.4641016	3.8637033				
7	4.4939592	10.0978347	14.5917939	14.5917939	10.0978347	4.4939592			
8	5.1258309	13.1370712	21.8461510	25.6883559	21.8461510	13.1370712	5.1258309		
9	5.7587705	16.5817187	31.1634375	41.9863857	41.9863857	31.1634375	16.5817187	5.7587705	
10	6.3924532	20.4317291	42.8020611	64.8823963	74.2334292	64.8823963	42.8020611	20.4317291	6.3924532

例1：设计一个满足下列指标BW型模拟低通滤波器

$$\omega_p = 0.1\pi, \quad \omega_s = 0.4\pi, \quad A_p \leq 1\text{dB}, \quad A_s \geq 10\text{dB}$$

解：

(1) 计算 N 和 Ω_c

$$N \geq \frac{\log_{10} \left(\frac{10^{0.1 A_p} - 1}{10^{0.1 A_s} - 1} \right)}{2 \log_{10} (\omega_p / \omega_s)} = 1.28$$

取 $N=2$ ，将 $N=2$ 代入满足通带的方程

$$\Omega_c = \frac{\omega_p}{(10^{0.1} - 1)^{1/4}} = 0.4404$$

(2) 确定滤波器的极点

$$s_k = \omega_c e^{j\pi(\frac{1}{2} + \frac{2k-1}{2N})}; k=1, 2, \dots, N$$

$$s_1 = \omega_c e^{j\frac{3}{4}\pi} = \omega_c \left(-\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2} \right) \quad s_2 = \Omega_c e^{j\frac{5}{4}\pi} = \Omega_c \left(-\frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2} \right)$$

(3) 确定模拟低通滤波器的系统函数 $H_L(s)$

$$H_L(s) = \prod_{k=1}^N \frac{\Omega_c}{s - s_k} = \frac{\Omega_c^2}{s^2 + \sqrt{2}\omega_c s + \omega_c^2} = \frac{0.4404^2}{s^2 + 0.6228s + 0.4404^2}$$

验证： $A_p=0.9999\text{db}$, 满足指标； $A_s= 18.2795 \text{ db}$ 超过指标

题：BW型模拟低通滤波器N=2，Wc=0.4404，问设计模拟低通滤波器系统函数

表 5 - 1 巴特沃思归一化低通滤波器分母多项式

$s^N + a_{N-1}s^{N-1} + \dots + a_2s^2 + a_1s + 1$ 的系数 ($a_1 = a_N = 1$)

N	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9
1	1								
2	1.4142436								
3	2.0000000	2.0000000							
4	2.6131259	3.4142136	2.6131259						
5	3.2360680	5.2360680	5.2360680	3.2360680					
6	3.8637033	7.4641016	9.1416202	7.4641016	3.8637033				
7	4.4939592	10.0978347	14.5917939	14.5917939	10.0978347	4.4939592			
8	5.1258309	13.1370712	21.8461510	25.6883559	21.8461510	13.1370712	5.1258309		
9	5.7587705	16.5817187	31.1634375	41.9863857	41.9863857	31.1634375	16.5817187	5.7587705	
10	6.3924532	20.4317291	42.8020611	64.8823963	74.2334292	64.8823963	42.8020611	20.4317291	6.3924532

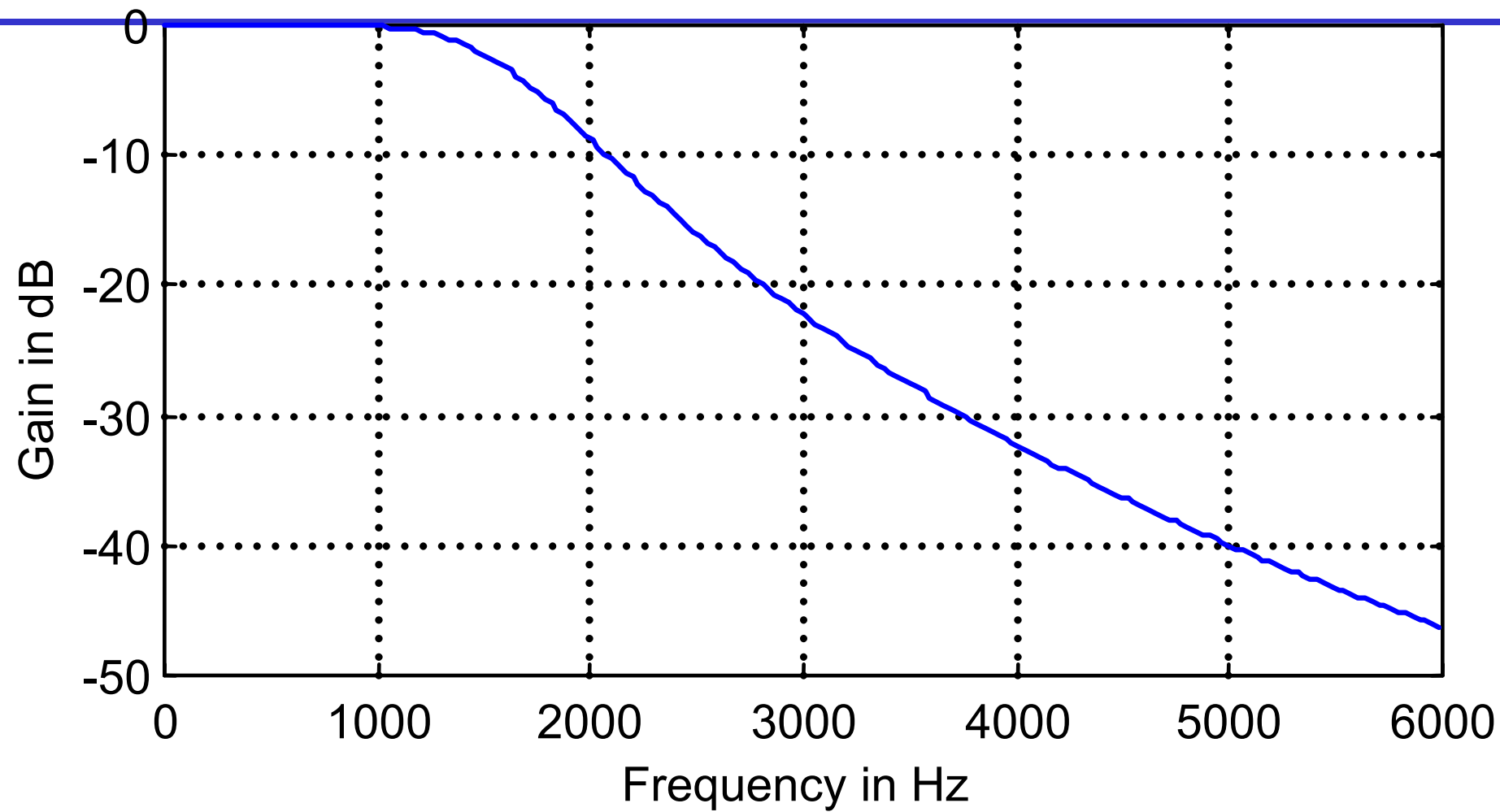
例：设计满足下列条件的模拟**Butterworth**低通滤波器

$$f_p=1\text{kHz}, \quad f_s=5\text{kHz}, \quad A_p=1\text{dB}, \quad A_s=40\text{dB}$$

```
Wp=2*pi*1000;Ws=2*pi*5000;Ap=1;As=40;
[N,Wc]=buttord(Wp,Ws,Ap,As,'s');
fprintf('Order of the filter=%.0f\n',N)
[num,den] = butter(N,Wc,'s');
disp('Numerator polynomial');
fprintf('%.4e\n',num);
disp('Denominator polynomial');
fprintf('%.4e\n',den);
omega=[Wp Ws];h = freqs(num,den,omega);
fprintf('Ap= %.4f\n',-20*log10(abs(h(1))));
fprintf('As= %.4f\n',-20*log10(abs(h(2))));
omega = [0: 200: 12000*pi];
h = freqs(num,den,omega);
gain = 20*log10(abs(h));plot (omega/(2*pi),gain);
xlabel('Frequency in Hz'); ylabel('Gain in dB');
```

例：设计满足下列条件的模拟**Butterworth**低通滤波器

$$f_p=1\text{kHz}, \quad f_s=5\text{kHz}, \quad A_p=1\text{dB}, \quad A_s=40\text{dB}$$



$N=4$ $A_p= 0.1098 \text{ dB}$ $A_s= 40.0000 \text{ dB}$

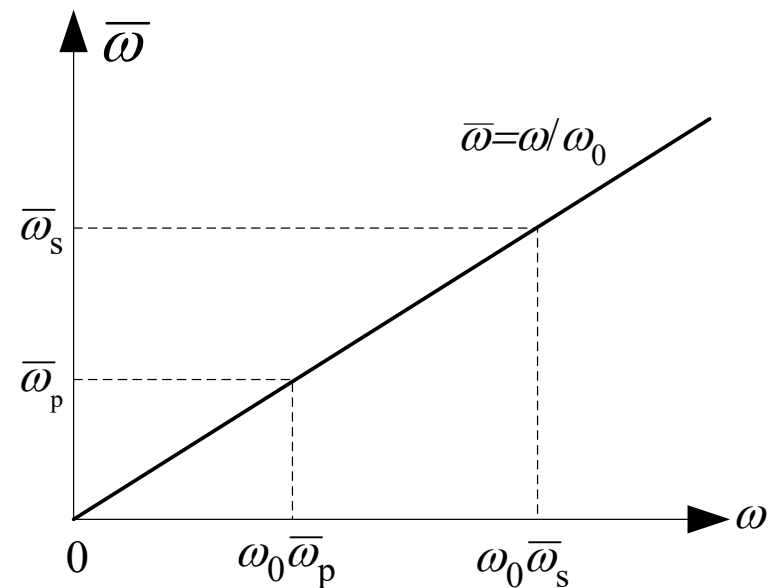
模拟频率变换

1. 原型低通HL到低通HLP的变换

$$H_{LP}(s) = H_L(\bar{s}) \Big|_{\bar{s}=s/\omega_0}$$

将 $\bar{s} = j\bar{\omega}$, $s = j\omega$ 代入 $\bar{s} = s / \omega_0$ 中得

$$\bar{\omega} = \omega / \omega_0$$



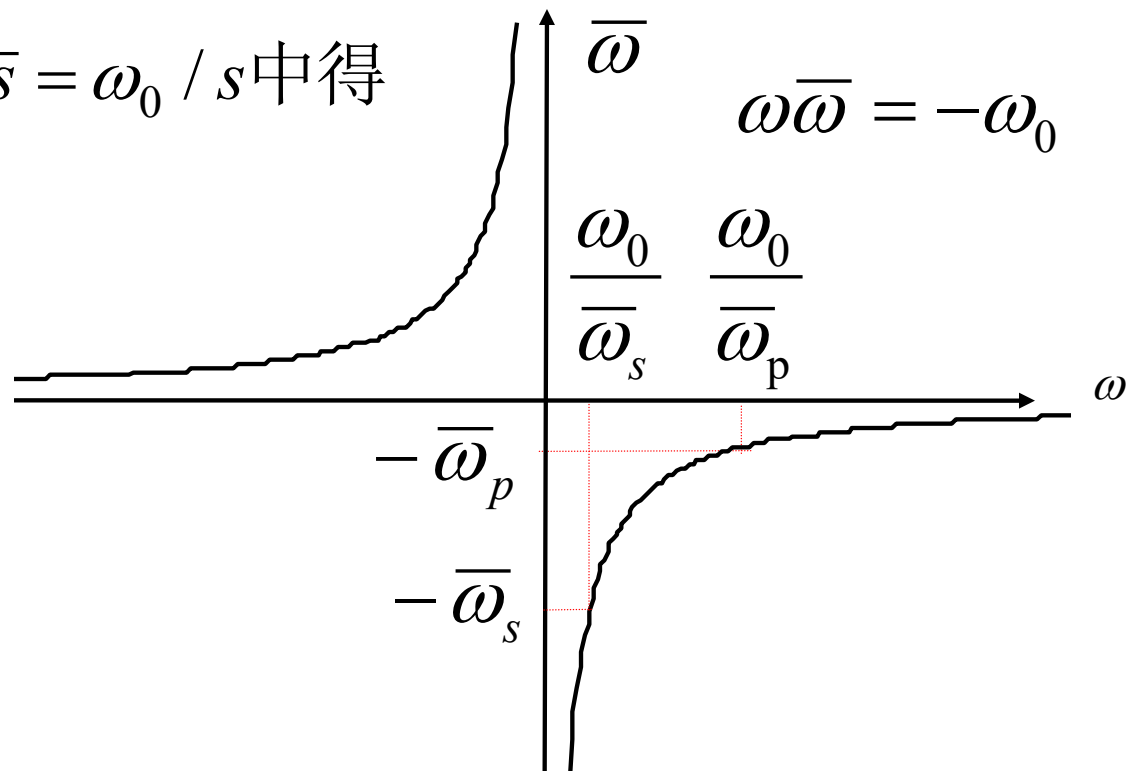
模拟频率变换

2. 原型低通到高通的变换

$$H_{\text{HP}}(s) = H_{\text{L}}(\bar{s}) \Big|_{\bar{s} = \omega_0 / s}$$

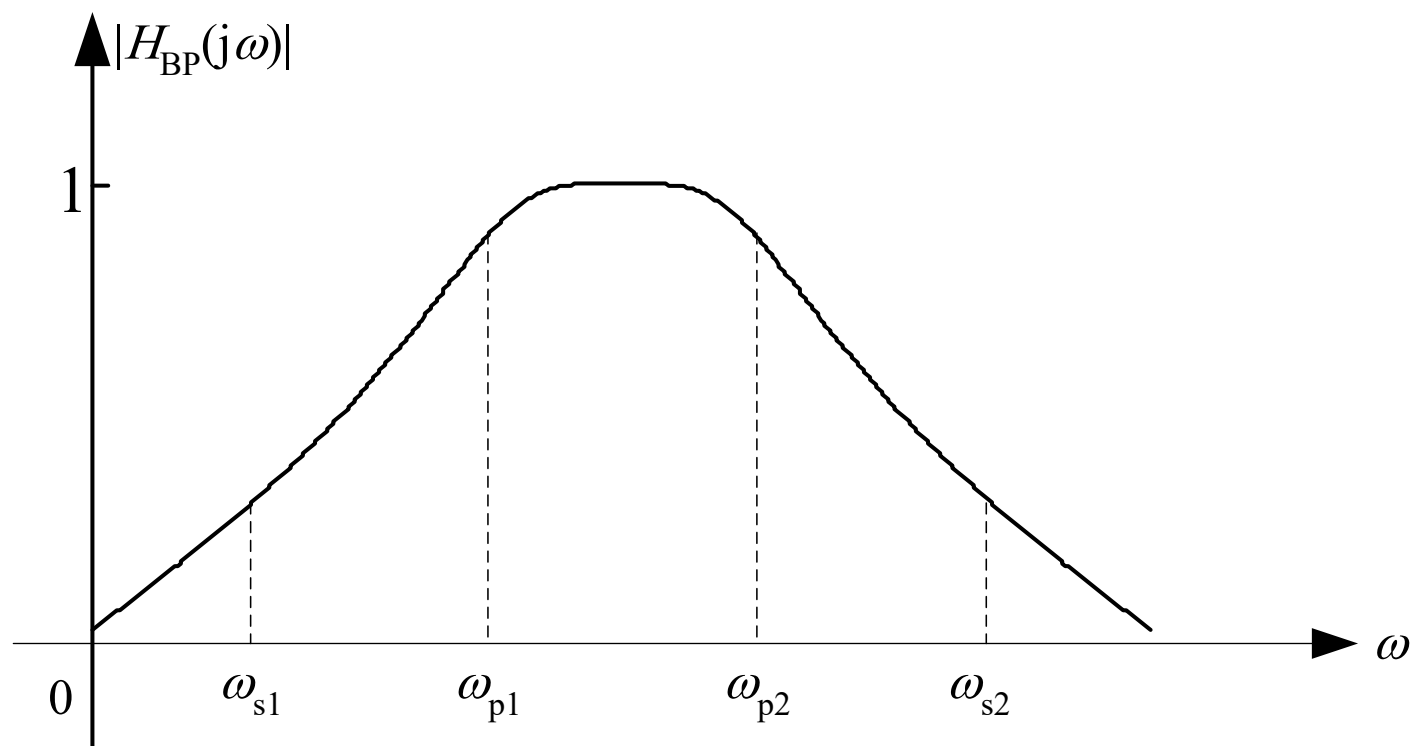
将 $\bar{s} = j\bar{\omega}$, $s = j\omega$ 代入 $\bar{s} = \omega_0 / s$ 中得

$$\omega \bar{\omega} = -\omega_0$$



模拟频率变换

3. 原型低通到带通的变换



模拟带通滤波器的幅度响应

模拟频率变换

3. 原型低通到带通的变换

$$H_{BP}(s) = H_L(\bar{s}) \bigg|_{\bar{s} = \frac{s^2 + \omega_0^2}{Bs}}$$

$$\text{其中: } B = \omega_{p2} - \omega_{p1} \quad \omega_0^2 = \omega_{p1} \omega_{p2}$$

$$\text{将 } \bar{s} = j\bar{\omega}, s = j\omega \text{ 代入 } \bar{s} = \frac{s^2 + \omega_0^2}{Bs} \text{ 中得}$$

$$\omega = 0.5(\bar{\omega}B \pm \sqrt{(\bar{\omega}B)^2 + 4\omega_0^2})$$

取+号时, 无论 $\bar{\omega}$ 怎样变化, ω 均为正

取-号时, 无论 $\bar{\omega}$ 怎样变化, ω 均为负

只讨论取+号
情况

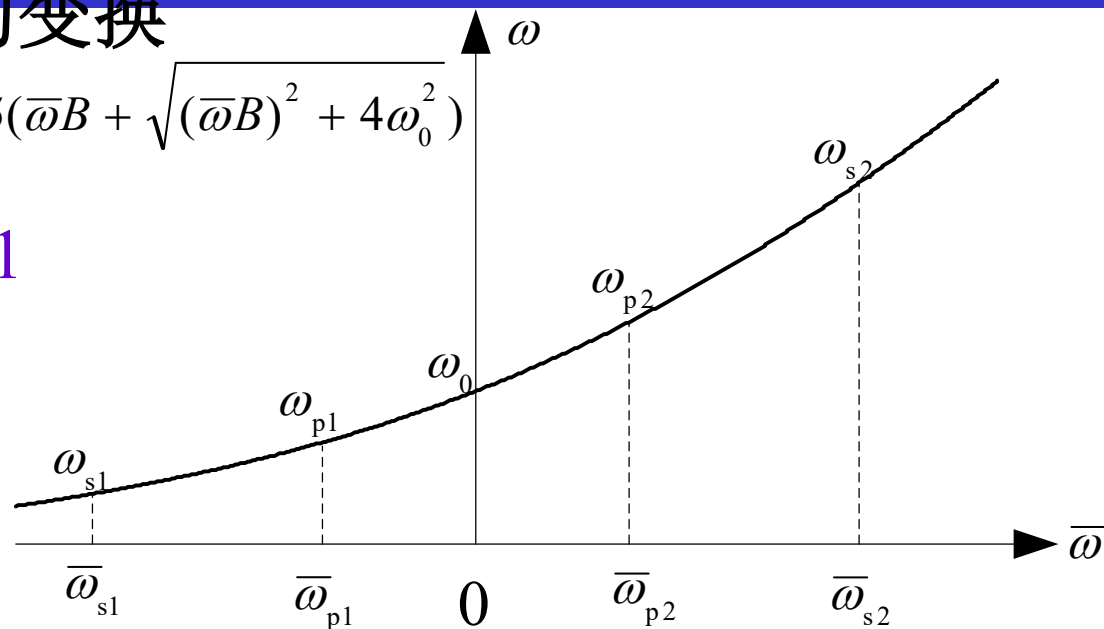
模拟频率变换

3. 原型低通到带通的变换

$$\omega = 0.5(\bar{\omega}B + \sqrt{(\bar{\omega}B)^2 + 4\omega_0^2})$$

$$\bar{\omega}_{p1} = \frac{\omega_{p1}^2 - \omega_{p1}\omega_{p2}}{(\omega_{p2} - \omega_{p1})\omega_{p1}} = -1$$

$$\bar{\omega}_{p2} = \frac{\omega_{p2}^2 - \omega_{p1}\omega_{p2}}{(\omega_{p2} - \omega_{p1})\omega_{p2}} = 1$$



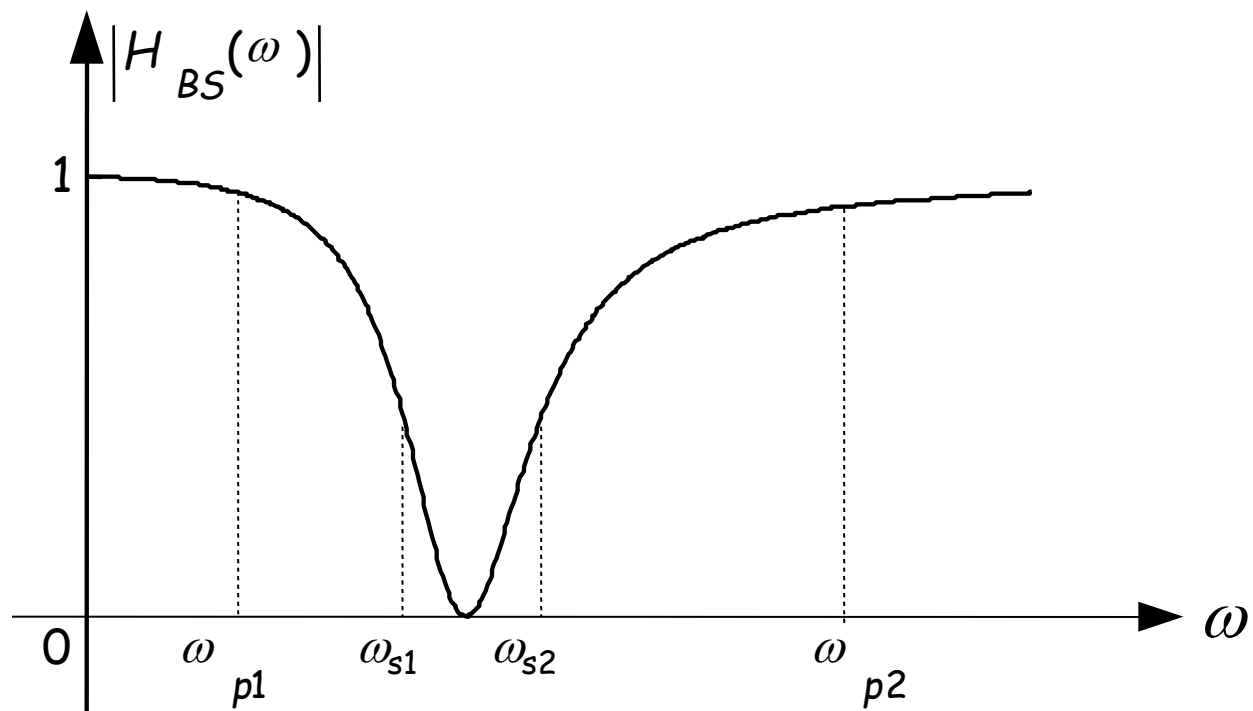
$$\bar{\omega}_{s1} = \frac{\omega_{s1}^2 - \omega_{s1}\omega_{s2}}{(\omega_{p2} - \omega_{p1})\omega_{s1}}$$

$$\bar{\omega}_{s2} = \frac{\omega_{s2}^2 - \omega_{s1}\omega_{s2}}{(\omega_{p2} - \omega_{p1})\omega_{s2}}$$

$$\bar{\omega}_s = \min\{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\}$$

模拟频率变换

4. 原型低通到带阻通的变换



模拟带阻滤波器的幅度响应

模拟频率变换

4. 原型低通到带阻通的变换

$$H_{BS}(s) = H_L(\bar{s}) \bigg|_{\bar{s} = \frac{Bs}{s^2 + \omega_0^2}}$$

$$\text{其中: } B = \omega_{s2} - \omega_{s1} \quad \omega_0^2 = \omega_{s1} \omega_{s2}$$

将 $\bar{s} = j\bar{\omega}, s = j\omega$ 代入 $\bar{s} = \frac{Bs}{s^2 + \omega_0^2}$ 中得

$$\omega = -\frac{B}{2\bar{\omega}} \pm \sqrt{\omega_0^2 + \left(\frac{B}{2\bar{\omega}}\right)^2}$$

模拟频率变换

4. 原型低通到带阻通的变换

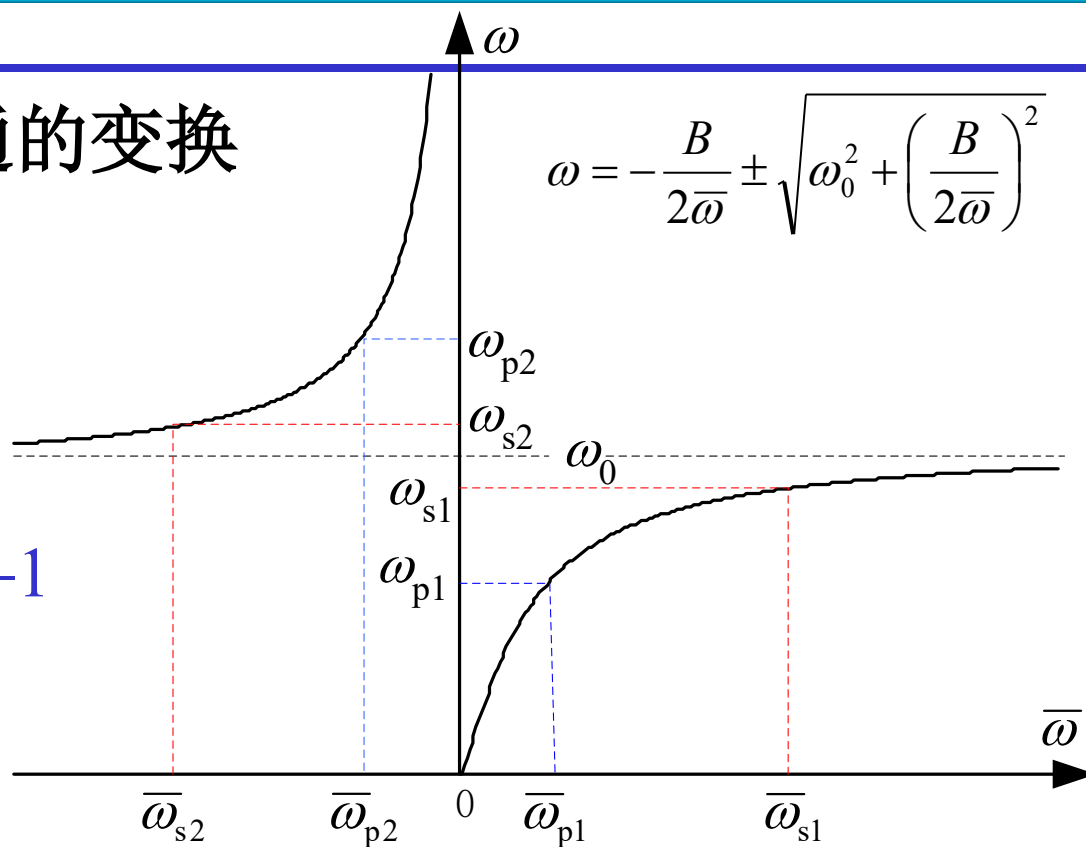
$$\bar{\omega}_{s1} = \frac{(\omega_{s2} - \omega_{s1})\omega_{s1}}{-\omega_{s1}^2 + \omega_{s1}\omega_{s2}} = 1$$

$$\bar{\omega}_{s2} = \frac{(\omega_{s2} - \omega_{s1})\omega_{s2}}{-\omega_{s2}^2 + \omega_{s1}\omega_{s2}} = -1$$

$$\bar{\omega}_{p1} = \frac{(\omega_{s2} - \omega_{s1})\omega_{p1}}{-\omega_{p1}^2 + \omega_{s1}\omega_{s2}}$$

$$\bar{\omega}_{p2} = \frac{(\omega_{s2} - \omega_{s1})\omega_{p2}}{-\omega_{p2}^2 + \omega_{s1}\omega_{s2}}$$

$$\bar{\omega}_p = \max\{|\bar{\omega}_{p1}|, |\bar{\omega}_{p2}|\}$$



模拟频率变换

变换类型	频率变换	复频率变换	注释
原型低通→低通	$\bar{\omega} = \omega / \omega_0$	$\bar{s} = s / \omega_0$	ω_0 是一正参数
原型低通→高通	$\bar{\omega} = \omega_0 / \omega$	$\bar{s} = \omega_0 / s$	ω_0 是一正参数
原型低通→带通	$\bar{\omega} = \frac{\omega^2 - \omega_0^2}{B\omega}$	$\bar{s} = \frac{s^2 - \omega_0^2}{Bs}$	$B = \omega_{p2} - \omega_{p1}$ $\omega_0^2 = \omega_{p1}\omega_{p2}$
原型低通→带阻	$\bar{\omega} = \frac{B\omega}{\omega^2 - \omega_0^2}$	$\bar{s} = \frac{Bs}{s^2 - \omega_0^2}$	$B = \omega_{s2} - \omega_{s1}$ $\omega_0^2 = \omega_{s1}\omega_{s2}$

模拟高通滤波器的设计

模拟高通滤波器的设计步骤:

(1)由高通滤波器的频率指标确定低通的频率指标

$$\overline{\omega}_p = 1 / \omega_p \quad \overline{\omega}_s = 1 / \omega_s$$

(2)设计满足指标 $\overline{\omega}_p, \overline{\omega}_s, A_p, A_s$ 的原型低通滤波器 $H_L(\bar{s})$

(3)由复频率变换将原型低通转换为高通 $H_{HP}(s)$

$$H_{HP}(s) = H_L(\bar{s}) \bigg|_{\bar{s} = \frac{1}{s}}$$

MATLAB实现

```
[numt,dent] = lp2hp(num,den,W0)
```

例1： 设计满足下列条件的模拟**BW**型高通滤波器
 $f_p=5\text{kHz}$, $f_s=1\text{kHz}$, $A_p\leq 1\text{dB}$, $A_s\geq 40\text{dB}$

解： **(1)**将高通指标转换为原型低通滤波器技术指标

$$\bar{\omega}_p = 1/(2\pi \times 5000), \bar{\omega}_s = 1/(2\pi \times 1000) , A_p \leq 1\text{dB}, A_s \geq 40\text{dB}$$

(2) 设计**BW**型原型低通滤波器

$$N \geq \frac{\log_{10} \left(\frac{10^{0.1A_p} - 1}{10^{0.1A_s} - 1} \right)}{2 \log_{10} (\omega_p / \omega_s)} = 3.28 \quad \text{取 } N=4$$

$$\omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s} - 1)^{1/2N}} = 5.033 \times 10^{-5}$$

例1：设计满足下列条件的模拟BW型高通滤波器

$$f_p=5\text{KHz}, \quad f_s=1\text{kHz}, \quad A_p \leq 1\text{dB}, \quad A_s \geq 40\text{dB}$$

(2) 设计BW型原型低通滤波器

$$N=4, \quad \omega_c=5.033 \times 10^{-5}$$

BW型原型低通滤波器的系统函数为

$$\begin{aligned} H_L(\bar{s}) &= \frac{1}{[(\bar{s} / \omega_c)^2 + 0.7654\bar{s} / \omega_c + 1][(\bar{s} / \omega_c)^2 + 1.8478\bar{s} / \omega_c + 1]} \\ &= \frac{6.4166 \times 10^{-18}}{\bar{s}^4 + 1.3152 \times 10^{-4} \bar{s}^3 + 8.6485 \times 10^{-9} \bar{s}^2 + 3.3315 \times 10^{-13} \bar{s} + 6.4166 \times 10^{-18}} \end{aligned}$$

(3) 由变换 $H_{HP}(s)=H_L(1/s)$ 获得高通滤波器

$$H_{HP}(s) = \frac{s^4}{s^4 + 5.192 \times 10^4 s^3 + 1.3478 \times 10^9 s^2 + 2.0497 \times 10^{13} s + 1.5585 \times 10^{17}}$$

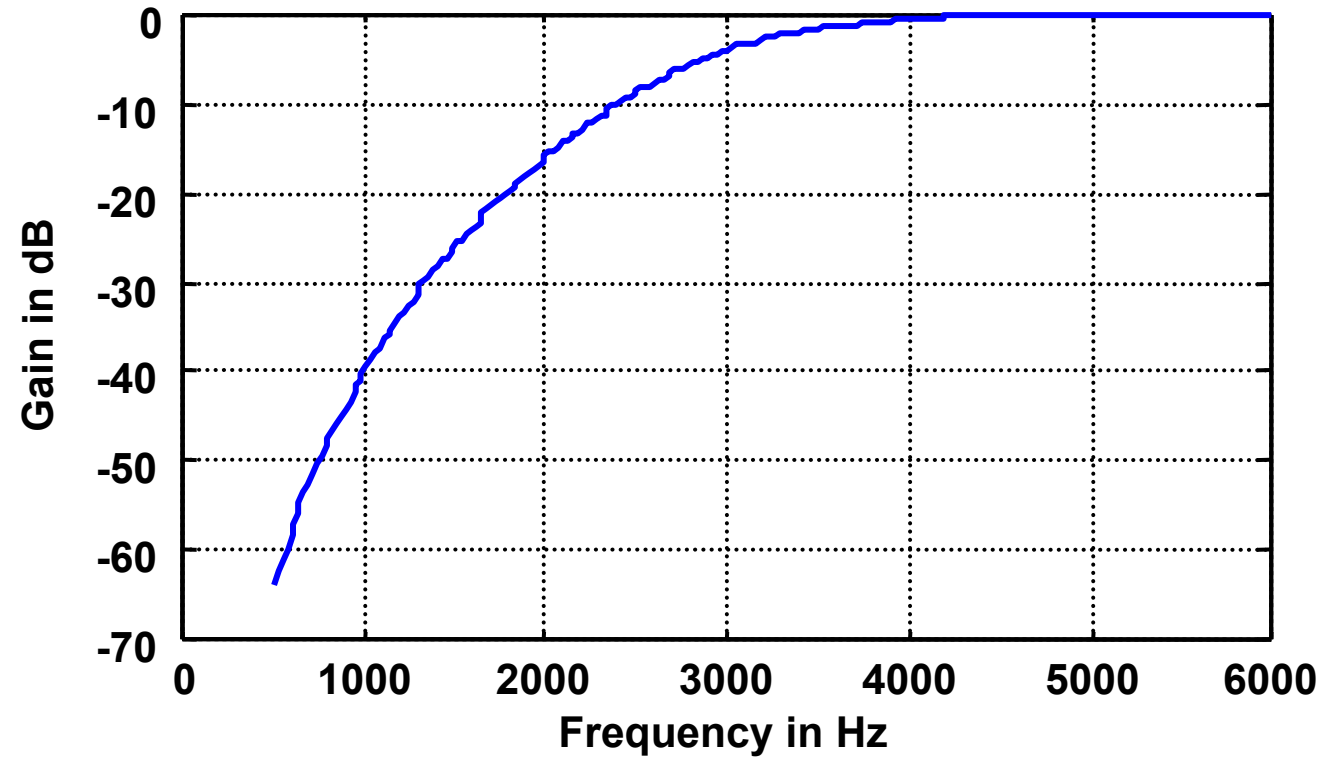
例1：设计满足下列条件的模拟BW型高通滤波器

$$f_p=5\text{KHz}, f_s=1\text{kHz}, A_p\leq 1\text{dB}, A_s\geq 40\text{dB}$$

%高通滤波器的设计

```
wp=1/(2*pi*5000);ws=1/(2*pi*1000);Ap=1;As=40;  
[N,Wc]=buttord(wp,ws,Ap,As,'s');  
[num,den] = butter(N,Wc,'s');  
disp('LP 分子多项式');  
fprintf('%.4e\n',num);  
disp('LP 分母多项式');  
fprintf('%.4e\n',den);  
[numt,dent] = lp2hp(num,den,1);  
disp('HP 分子多项式');  
fprintf('%.4e\n',numt);  
disp('HP 分母多项式');  
fprintf('%.4e\n',dent);
```

例1: 设计满足下列条件的模拟BW型高通滤波器
 $f_p=5\text{KHz}$, $f_s=1\text{kHz}$, $A_p\leq 1\text{dB}$, $A_s\geq 40\text{dB}$



$A_p= 40.0000$ $A_s= 0.1098$

模拟带通滤波器的设计

模拟带通滤波器的设计步骤:

(1)由带通滤波器的上下截频确定变换式中的参数

$$B = \omega_{p2} - \omega_{p1} \quad \omega_0^2 = \omega_{p1} \omega_{p2}$$

(2) 确定原型低通滤波器的阻带截频

$$\bar{\omega}_s = \min \{ |\bar{\omega}_{s1}|, |\bar{\omega}_{s2}| \}$$

其中

$$\bar{\omega}_{s1} = \frac{\omega_{s1}^2 - \omega_{p1} \omega_{p2}}{(\omega_{p2} - \omega_{p1}) \omega_{s1}} \quad \bar{\omega}_{s2} = \frac{\omega_{s2}^2 - \omega_{p1} \omega_{p2}}{(\omega_{p2} - \omega_{p1}) \omega_{s2}}$$

(3)设计通带截频为1 (rad/s)、阻带截频为 $\bar{\omega}_s$ 、通带衰减为 A_p dB、阻带衰减为 A_s dB的原型低通滤波器

模拟带通滤波器的设计

模拟带通滤波器的设计步骤:

(4) 将原型低通滤波器转换为带通滤波器 $H_{BP}(s)$

$$H_{BP}(s) = H_L(\bar{s}) \bigg|_{\bar{s} = \frac{s^2 + \omega_0^2}{Bs}}$$

MATLAB实现

```
[numt,dent] = lp2bp(num,den,W0,B)
```

例2：试设计一个满足下列指标的BW型带通滤波器

$$\omega_{p1}=6 \text{ rad/s}, \omega_{p2}=8 \text{ rad/s}, \omega_{s1}=4 \text{ rad/s}, \\ \omega_{s2}=11 \text{ rad/s}, A_p \leq 1 \text{ dB}, A_s \geq 32\text{dB}.$$

解：

(1)由带通滤波器的上下截频确定变换式中的参数

$$B = \omega_{p2} - \omega_{p1} = 2 \quad \omega_0^2 = \omega_{p1} \omega_{p2} = 48$$

(2) 确定原型低通滤波器的阻带截频

$$\bar{\omega}_{s1} = \frac{\omega_{s1}^2 - \omega_{p1} \omega_{p2}}{(\omega_{p2} - \omega_{p1}) \omega_{s1}} = -4 \quad \bar{\omega}_{s2} = \frac{\omega_{s2}^2 - \omega_{p1} \omega_{p2}}{(\omega_{p2} - \omega_{p1}) \omega_{s2}} = 3.3182$$

故
$$\bar{\omega}_s = \min \{|\bar{\omega}_{s1}|, |\bar{\omega}_{s2}|\} = 3.3182$$

例2：试设计一个满足下列指标的**BW**型带通滤波器

$$\omega_{p1}=6 \text{ rad/s}, \omega_{p2}=8 \text{ rad/s}, \omega_{s1}=4 \text{ rad/s}, \\ \omega_{s2}=11 \text{ rad/s}, A_p \leq 1 \text{ dB}, A_s \geq 32 \text{ dB}.$$

解：(3) 设计满足下列指标的原型低通滤波器

$$\bar{\omega}_p = 1, A_p = 1 \text{ dB}, \bar{\omega}_s = 3.3182, A_s = 32 \text{ dB}$$

$$N \geq \frac{\log_{10} \left(\frac{10^{0.1A_p} - 1}{10^{0.1A_s} - 1} \right)}{2 \log_{10} (\bar{\omega}_p / \bar{\omega}_s)} = 3.6346 \quad \text{取 } N=4$$

$$\omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s} - 1)^{1/2N}} = 1.3211$$

例2：试设计一个满足下列指标的BW型带通滤波器

$$\omega_{p1}=6 \text{ rad/s}, \omega_{p2}=8 \text{ rad/s}, \omega_{s1}=4 \text{ rad/s}, \\ \omega_{s2}=11 \text{ rad/s},$$

$$A_p \leq 1 \text{ dB}, A_s \geq 32 \text{ dB}.$$

解：(3) 设计BW型原型低通滤波器 $N=4$, $\omega_c=1.3211$

$$H_L(\bar{s}) = \frac{1}{[(\bar{s} / \omega_c)^2 + 0.7654\bar{s} / \omega_c + 1][(\bar{s} / \omega_c)^2 + 1.8478\bar{s} / \omega_c + 1]}$$

4) 将原型低通滤波器转换为带通滤波器 $H_{BP}(s)$

$$H_{BP}(s) = H_L(\bar{s}) \bigg|_{\bar{s} = \frac{s^2 + \omega_0^2}{Bs}} \\ = \frac{48.7372s^4}{(s^4 + 2s^3 + 0.103 \times 10^3 s^2 + 0.0971 \times 10^3 s + 2.304 \times 10^3)} \times \\ \frac{1}{(s^4 + 0.0049 \times 10^3 s^3 + 0.103 \times 10^3 s^2 + 0.2343 \times 10^3 s + 2.304 \times 10^3)}$$



例2：试设计一个满足下列指标的BW型带通滤波器

$\omega_{p1}=6$ rad/s, $\omega_{p2}=8$ rad/s, $\omega_{s1}=4$ rad/s,
 $\omega_{s2}=11$ rad/s,

$A_p \leq 1$ dB, $A_s \geq 32$ dB.

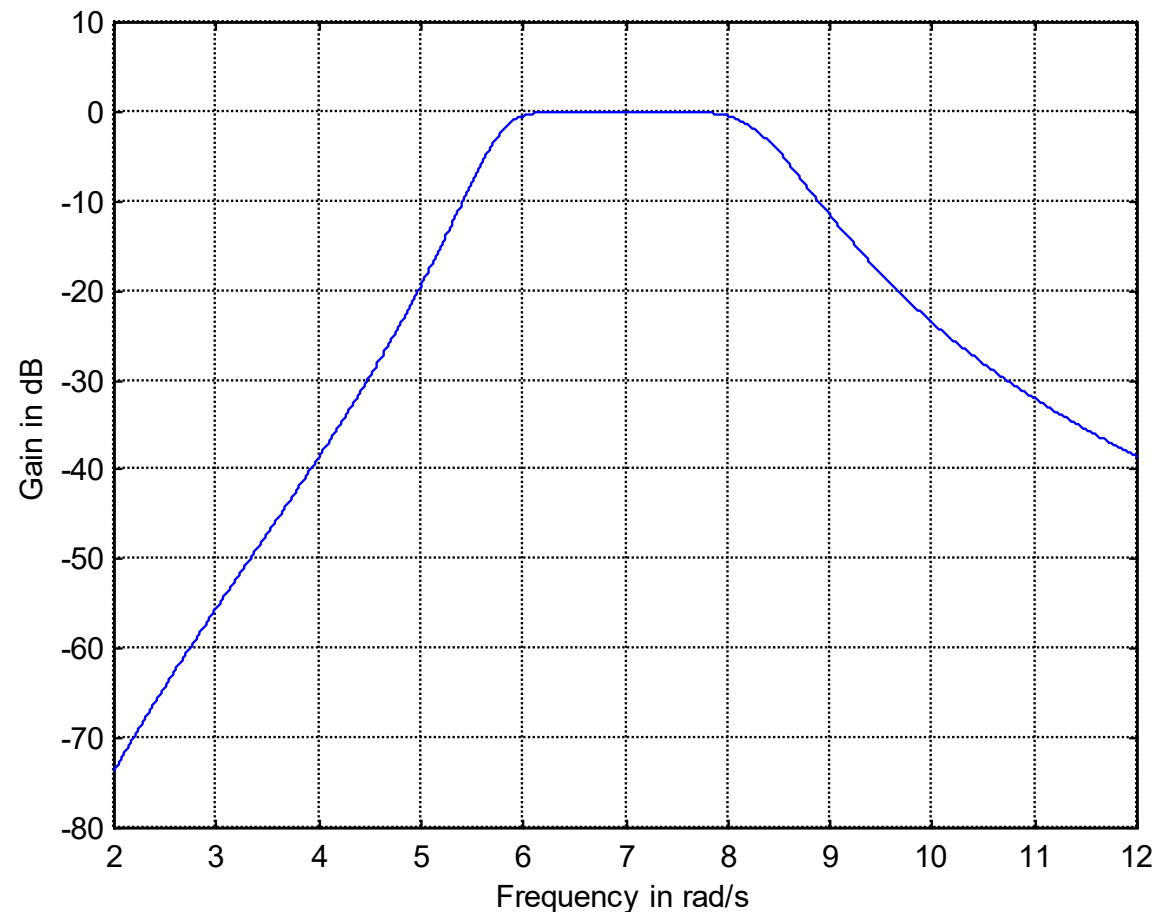
%带通滤波器的设计

```
wp=1;ws=3.3182;Ap=1;As=32;  
w0=sqrt(48);B=2;  
[N,Wc]=buttord(wp,ws,Ap,As,'s');  
[num,den] = butter(N,Wc,'s');  
[numt,dent] = lp2bp(num,den,w0,B);  
w=linspace(2,12,1000);  
h=freqs(numt,dent,w);  
plot(w,20*log10(abs(h))) ; grid ;  
xlabel('Frequency in rad/s');  
ylabel('Gain in dB')
```

例2：试设计一个满足下列指标的**BW**型带通滤波器

$$\omega_{p1}=6 \text{ rad/s}, \omega_{p2}=8 \text{ rad/s}, \omega_{s1}=4 \text{ rad/s}, \\ \omega_{s2}=11 \text{ rad/s},$$

$$A_p \leq 1 \text{ dB}, A_s \geq 32 \text{ dB}.$$



模拟带阻滤波器的设计

模拟带阻滤波器的设计步骤:

(1)由带阻滤波器通带的上下截频确定变换式中的参数

$$B = \omega_{s2} - \omega_{s1} \quad \omega_0^2 = \omega_{s1} \omega_{s2}$$

(2)确定原型低通滤波器的通带截频

$$\overline{\omega}_p = \max \{ |\overline{\omega}_{p1}|, |\overline{\omega}_{p2}| \}$$

其中

$$\overline{\omega}_{p1} = \frac{(\omega_{s2} - \omega_{s1})\omega_{p1}}{-\omega_{p1}^2 + \omega_{s1}\omega_{s2}} \quad \overline{\omega}_{p2} = \frac{(\omega_{s2} - \omega_{s1})\omega_{p2}}{-\omega_{p2}^2 + \omega_{s1}\omega_{s2}}$$

(3)设计通带截频为 $\overline{\omega}_p$ 、阻带截频为1 (rad/s)、通带衰减为 A_p dB、通带衰减为 A_s dB的低通滤波器

模拟带阻滤波器的设计

模拟带阻滤波器的设计步骤:

(4) 将低通滤波器转换为带阻滤波器 $H_{BS}(s)$

$$H_{BS}(s) = H_L(\bar{s}) \left| \bar{s} = \frac{Bs}{s^2 + \omega_0^2} \right.$$

MATLAB实现

```
[numt,dent] = lp2bs(num,den,W0,B)
```

例3：试设计一个满足下列指标的BW型带阻滤波器

$$A_p=1\text{dB}; A_s=20\text{dB}; \omega_{p1}=10; \omega_{p2}=30; \omega_{s1}=19; \omega_{s2}=21$$

解：

(1)由带阻滤波器通带的上下截频确定变换式中的参数

$$B = \omega_{s2} - \omega_{s1} = 2 \quad \omega_0^2 = \omega_{s1} \omega_{s2} = 399$$

(2)确定原型低通滤波器的通带截频

$$\overline{\omega}_{p1} = \frac{(\omega_{s2} - \omega_{s1})\omega_{p1}}{-\omega_{p1}^2 + \omega_{s1}\omega_{s2}} = 0.0669 \quad \overline{\omega}_{p2} = \frac{(\omega_{s2} - \omega_{s1})\omega_{p2}}{-\omega_{p2}^2 + \omega_{s1}\omega_{s2}} = -0.1198$$

$$\text{故} \quad \overline{\omega}_p = \max \{|\overline{\omega}_{p1}|, |\overline{\omega}_{p2}|\} = 0.1198$$

例3： 试设计一个满足下列指标的**BW**型带阻滤波器

$$\underline{A_p=1\text{dB}; A_s=20\text{dB}; \omega_{p1}=10; \omega_{p2}=30; \omega_{s1}=19; \omega_{s2}=21}$$

解： (3)设计满足下列指标的**BW**原型低通滤波器

$$\bar{\omega}_p = 0.1198, A_p = 1\text{dB}, \bar{\omega}_s = 1, A_s = 20\text{dB}$$

$$N \geq \frac{\log_{10} \left(\frac{10^{0.1A_p} - 1}{10^{0.1A_s} - 1} \right)}{2 \log_{10} (\bar{\omega}_p / \bar{\omega}_s)} = 1.4009 \quad \text{取 } N=2$$

$$\omega_c = \frac{\bar{\omega}_s}{(10^{0.1A_s} - 1)^{1/2N}} = 0.3170$$

例3：试设计一个满足下列指标的**BW**型带阻滤波器

$$A_p=1\text{dB}; A_s=20\text{dB}; \omega_{p1}=10; \omega_{p2}=30; \omega_{s1}=19; \omega_{s2}=21$$

解： (3) 设计**BW**型原型低通滤波器 $N=2$, $\omega_c=0.3170$

$$H_L(\bar{s}) = \frac{1}{(\bar{s} / \omega_c)^2 + \sqrt{2}\bar{s} / \omega_c + 1} = \frac{0.1005}{\bar{s}^2 + 0.4483\bar{s} + 0.1005}$$

4) 将原型低通滤波器转换为带阻滤波器 $H_{BS}(s)$

$$H_{BS}(s) = H_L(\bar{s}) \bigg|_{\bar{s} = \frac{Bs}{s^2 + \omega_0^2}} = \frac{s^4 + 7.98 \times 10^2 s^2 + 1.5920 \times 10^5}{s^4 + 8.9518 s^3 + 8.3780 \times 10^2 s^2 + 3.5718 \times 10^3 s + 1.5920 \times 10^5}$$

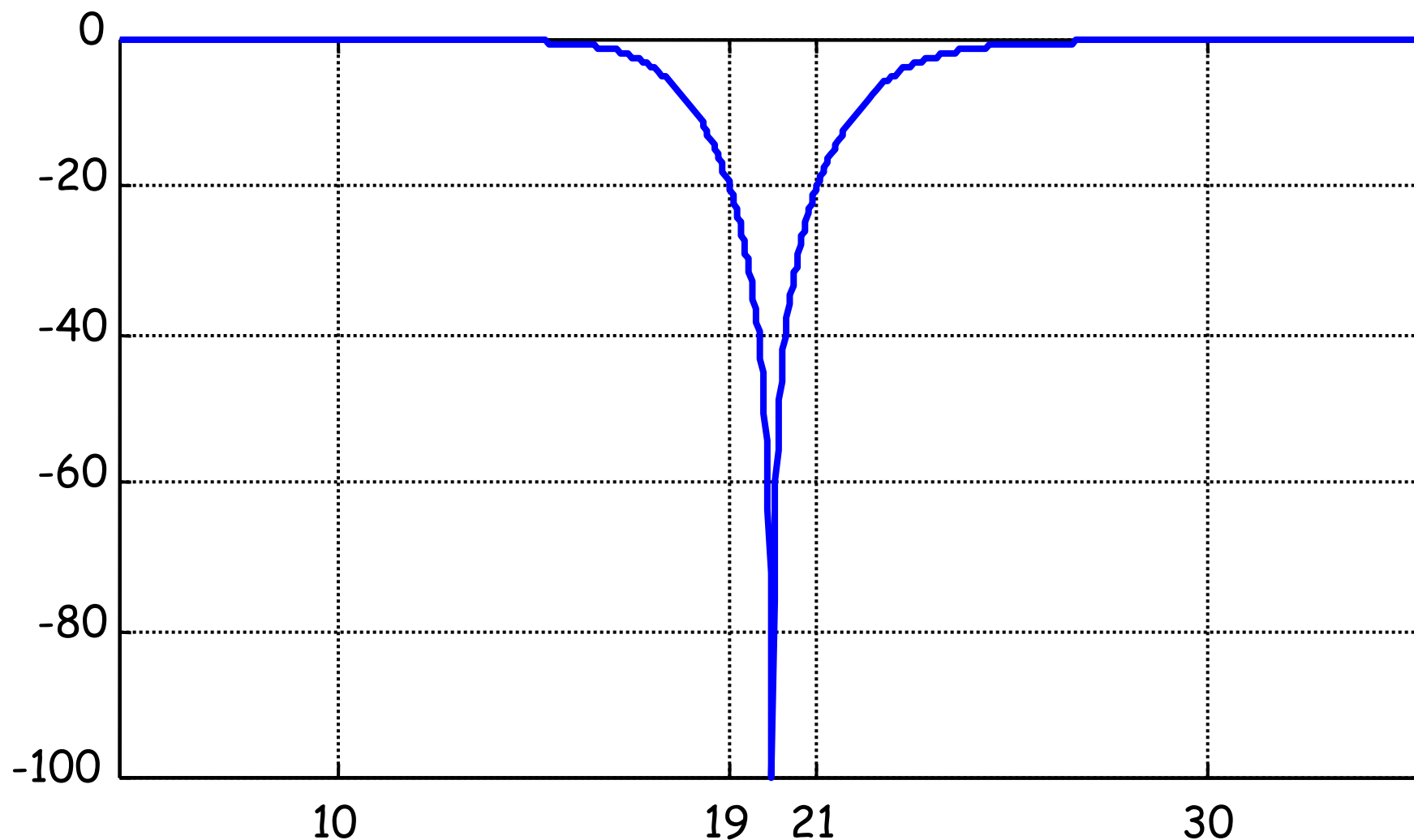
例3： 试设计一个满足下列指标的BW型带阻滤波器

$$A_p=1\text{dB}; A_s=20\text{dB}; \omega_{p1}=10; \omega_{p2}=30; \omega_{s1}=19; \omega_{s2}=21。$$

```
Ap=1;As=20;wp1=10;wp2=30;ws1=19;ws2=21;
B=ws2-ws1;w0=sqrt(ws1*ws2);
wLp1=B*wp1/(w0*w0-wp1*wp1);
wLp2=B*wp2/(w0*w0-wp2*wp2);
wLp=max(abs(wLp1),abs(wLp2));
[N,Wc]=buttord(wLp,1,Ap,As,'s');
[num,den] = butter(N,Wc,'s');
[numt,dent]=lp2bs(num,den,w0,B);
w=linspace(5,35,1000);
h=freqs(numt,dent,w);
plot(w,20*log10(abs(h)));
w=[wp1 ws1 ws2 wp2];
set(gca,'xtick',w);grid;
h=freqs(numt,dent,w);A=-20*log10(abs(h))
```

例3： 试设计一个满足下列指标的BW型带阻滤波器

$$A_p=1\text{dB}; A_s=20\text{dB}; \omega_{p1}=10; \omega_{p2}=30; \omega_{s1}=19; \omega_{s2}=21。$$





6.4 脉冲响应不变法设计IIR数字滤波器

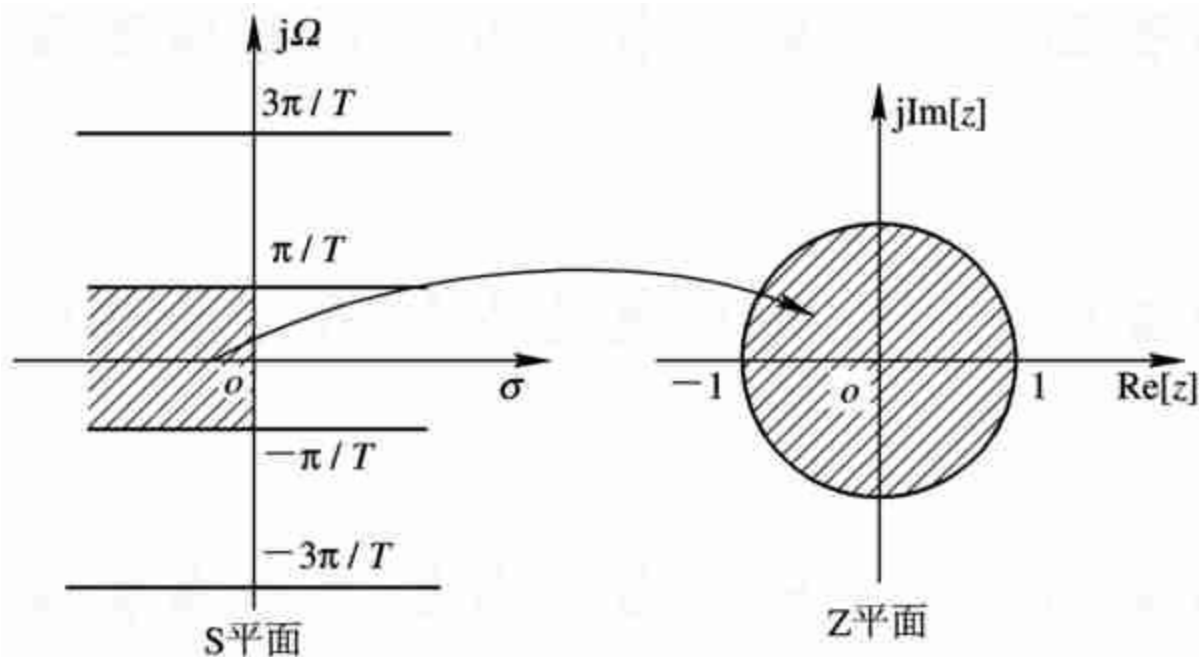
变换原理

模拟滤波器的冲激响应 $h_a(t)$ 进行等间隔采样，使 $h(n)$ 正好等于 $h_a(t)$ 的采样值，满足

$$h(n) = h_a(nT)$$

式中， T 是采样周期。

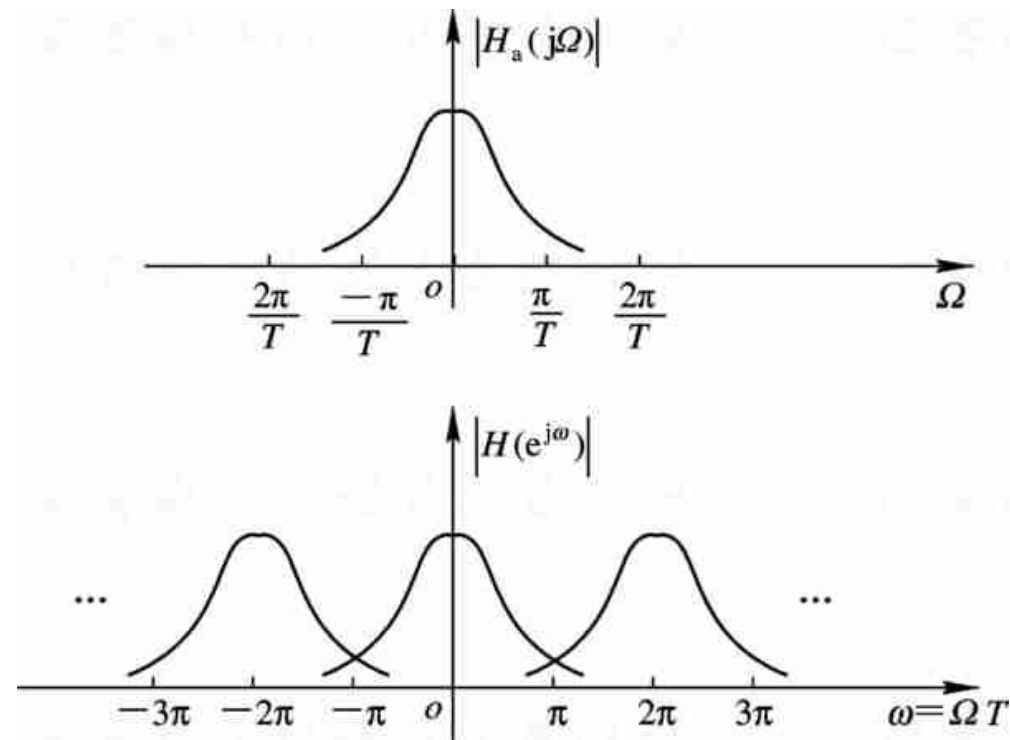
$$z = e^{sT}$$





混叠失真

$$H(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} H_a\left(j\frac{\omega - 2\pi k}{T}\right)$$





模拟滤波器的数字化方法

$$H_a(s) \xrightarrow{\text{green arrow}} h_a(t) \xrightarrow{\text{green arrow}} h(n)=h_a(nT) \xrightarrow{\text{green arrow}} H(z)$$

设模拟滤波器的系统函数 $H_a(s)$ 只有单阶极点,

因此可将

$$H_a(s) = \sum_{k=1}^N \frac{A_k}{s - s_k} \xrightarrow{\text{green arrow}} h_a(t) = F^{-1}[H_a(s)] = \sum_{k=1}^N A_k e^{s_k t} u(t)$$

$$h(n) = h_a(nT) = \sum_{k=1}^N A_k e^{s_k nT} u(n) = \sum_{k=1}^N A_k (e^{s_k T})^n u(n)$$

$$\begin{aligned} h(z) &= \sum_{n=-\infty}^{\infty} h(n) z^{-n} = \sum_{n=0}^{\infty} \sum_{k=1}^N A_k (e^{s_k T} z^{-1})^n = \sum_{k=1}^N A_k \sum_{n=0}^{\infty} (e^{s_k T} z^{-1})^n \\ &= \sum_{k=1}^N \frac{A_k}{1 - e^{s_k T} z^{-1}} \end{aligned}$$



模拟滤波器的数字化方法

从前式看，数字滤波器频率响应还与采样间隔 T 成反比：

$$H(e^{j\omega}) = \frac{1}{T} H_a\left(j \frac{\omega}{T}\right) \quad |\omega| < \pi$$

如果采样频率很高，即 T 很小，数字滤波器可能具有太高的增益，这是不希望的。为了使数字滤波器增益不随采样频率而变化，可以作以下简单的修正，令

$$h(n) = T h_a(nT)$$

则有：

$$H(z) = \sum_{k=1}^N \frac{T A_k}{1 - e^{s_k T} z^{-1}}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_a\left(j \frac{\omega}{T} - j \frac{2\pi}{T} k\right) \approx H_a\left(j \frac{\omega}{T}\right)$$